

Mah Hao Jun

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EDUCATION

National University of Singapore

Aug. 2024 – May 2028

Bachelor of Computing, Computer Science

Singapore

Cumulative GPA: 4.5

Relevant Coursework: Software Engineering; Data Structures and Algorithms; Programming Methodology

EXPERIENCE

Citi

Jun. 2026 – Aug. 2026

Software Engineer Intern

Singapore

- Built a Grafana observability dashboard with alerting for a core banking platform, surfacing **real-time metrics** across **100,000+ daily corporate account transactions** to support operational monitoring.
- Refactored a **Spring Boot (Java)** microservice codebase into **shared internal libraries**, eliminating code duplication across services and reducing estimated development time for future microservices by **15%**.
- Developed an end-to-end **Kafka** test suite that injects synthetic transactions and validates downstream service output, ensuring correctness across critical **event-driven processing pipelines**.

Fly Fairly

Jan. 2026 – Apr. 2026

Software Engineer Intern

Singapore

- Identified and eliminated **3 key operational bottlenecks** through structured stakeholder interviews.
- Built a **Next.js** back-office reporting dashboard that centralized project metrics and reporting workflows, reducing manual cross-team status coordination and improving estimated sprint throughput by **20%**.
- Influenced technical roadmaps for **3 of 6** departments by documenting modernization proposals for stakeholder-identified workflow problems, including recommended architecture, implementation tradeoffs, and rollout plans.

Toppan Ecquaria

May 2025 – Jul. 2025

Software Engineer Intern

Singapore

- Integrated **RAG pipelines** and **LLM agents** into enterprise SaaS products, enabling context-aware document Q&A across **10+ client deployments** and supporting new product adoption and upsell opportunities.
- Enhanced a multilingual AI video dubbing product licensed at **\$5,000 per deployment** by integrating **Kokoro** audio generation and **LatentSync** lip-sync models to improve automated speech-to-video alignment.
- Modernized **React** and **FastAPI** web applications used across Toppan Next Group by refactoring frontend components and backend APIs, improving maintainability and production readiness for a **2,000-user** deployment environment.

PROJECTS

GroceryPicker | Independent Software Development Project (Orbital – Apollo 11 Level)

May 2025 – Jul. 2025

- Built an Android application with **React Native + Express.js** that returns the best prices for each grocery out of every major supermarket, allowing users to minimize spending with ease.
- Implemented rigorous and scheduled web scraping of supermarket groceries with **Playwright** hosted on **GCP** to obtain a database of **50k groceries**, which is stored in PostgreSQL (Supabase).
- Implemented retrieval-augmented semantic search over the **50k grocery records**, enabling users to query products by natural language and rank matching items by price.

Aimlabs & Parkour

May 2024 – Jul. 2024

- Engineered a custom **3D game engine** in **C++** and **OpenGL** from scratch, implementing a scene graph, frustum culling, and a custom memory allocator to achieve stable **60+ FPS** rendering on mid-range hardware.

EXTRACURRICULAR ACTIVITIES

Google Developer Student Club @ NUS

Aug. 2025 – Present

Software Engineer

Singapore

- Building a **Next.js + FastAPI** volunteer platform with **React** and **TanStack Query** for real-time workflows.

NUS Fintech Society

Aug. 2025 – Present

Software Engineer

Singapore

- Building an **AI-powered itinerary planner** with HeyMax using social media transcription pipelines.

SKILLS

Languages : Python, JavaScript, TypeScript, Java

Frontend & Backend : React, React Native, Next.js, FastAPI, Express.js

Databases & Cloud : PostgreSQL, Supabase, Google Cloud Platform

Events & Programmes : Jane Street PORTAL